Modélisation, estimation, simulation des risques climatiques
Session 6

IAM Controversies

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Ecole polytechnique

IAMs in practice: evaluating the social cost of carbon

- IAMs are not only academic tools: since 2010, the US federal government uses them to compute the **social cost of carbon** (SCC), the damage caused by one extra ton of CO_2.
- An interagency working group averages three IAMs (DICE, PAGE, FUND); revisions across administrations show how normative choices move the number.

Administration	SCC ($/tCO$_2$)	Main change
Obama (2010)	21	IWG, 3% discount rate
Obama (2016)	42	updated DICE/PAGE/FUND runs
Trump (2017–2021)	1–7	domestic damages, 7% rate
Biden (2021)	51	interim, global damages
EPA (2023)	190	updated science, Ramsey 2%

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How relevant are IAM models?

- **IAM-believiers:** they are relevant at least on normative grounds to provide discussion basis on mitigation scenarios.

- **IAM-deniers:** they are useless to guide policy scenarios → the Pindyck (2013, 2017) critique enumerates a list of flaws:

  1. "*certain inputs [parameters] are arbitrary, but have huge effects on the SCC estimates the models produce*";
  2. "*the models' descriptions of the impact of climate change are completely ad hoc, with no theoretical or empirical foundation*";
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Two debates behind the critiques of IAMs

- IAMs leave many degrees of freedom: the same framework can support very different climate policies → an impression of lack of discipline.

- Two parameters illustrate this paradox:

  1. **Discounting:** the Stern–Nordhaus debate. How much weight should future damages receive relative to mitigation costs today?

  2. **Damage functions:** still an ongoing debate, on two fronts (treated in the sections below):

     2.1 **functional form:** how large are climate damages, and how should abrupt, irreversible losses be represented?

     2.2 **temperature elasticity:** how fast do damages rise with warming — quadratic in T, or higher-order terms?

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- In 2006, the main view on climate change among economists was the Nordhaus (1992)'s view: optimal to warm up to $3.5°C$ the planet.

- The Stern (2006)'s view shook this general agreement on many aspects:
  - the core argument: the price of inaction would be extraordinary and the cost of action modest ;
  - irreversible impacts from climate change threatening access to water, food production, health, and use of land and the environment;
  - IAM supports such actions in particular by changing discount factor → debate.



Nordhaus Stern

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The Discount Factor Controversy

- What's discount factor? Welfare function as a discounted sum of future gains:

$$\mathcal{W}_t = L_t U(c_t) + \beta_\Delta \mathcal{W}_{t+\Delta}$$

- β_Δ is a reduced-form parameter: a single number summarizing several distinct motives for weighting the future:
  1. psychological aspects (time-preference $\rightarrow \beta_\Delta$ low),
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The Discount Factor Controversy

- In practice, β_Δ not observable, but can be inferred from financial markets.
- Introducing a one-period financial claim:

$$\max_{\{c_t, b_t\}} \sum_{i=0}^{\infty} \beta_\Delta^i \frac{c_{t+\Delta i}^{1-\gamma}}{1-\gamma} + \lambda_t \ [b_t (1 + r_{t-\Delta}) - c_t - b_t]$$

$$\beta \frac{c_{t+\Delta}^{-\gamma}}{c_t^{-\gamma}} (1 + r_t) = 1 \rightarrow \ln \beta_\Delta = \gamma \ln g + \ln r$$

- By observing consumption growth and real interest rate, we can pin down β_Δ;
- But definition of relevant interest rate not clear: nominal bonds? corporate bonds?

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The Discount Factor Controversy (cont.)

- Nordhaus picks the market return on capital (corporate rates, about 3% real) and calibrates $\rho = 1.5\%$, i.e. $\beta_{\Delta=1} = 1/1.015 \simeq 0.985$;
- Stern (2006) argues that discount factor should be $\boxed{\beta_{\Delta=1} = 0.999}$ (versus 0.9852 in Nordhaus).

Nordhausś calibration of ρ across DICE vintages:

	ρ	$\beta_{\Delta=1} = 1/(1+\rho)$
DICE-1994	3.0%	0.9709
DICE-2007–2016	1.5%	0.9852
DICE-2023	1.0%	0.9901
Stern (2006), for contrast	0.1%	0.9990

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The discount factor

- The Stern (2006)'s report provides a scientific motivation for the Paris-Agreement.
- How? Consider FOC on T:

$$\begin{aligned}\lambda_{4,t} &= \beta_\Delta \left[\Phi_T^\top \lambda_{4,t+\Delta} + \lambda_{2,t+\Delta} s_{t+\Delta} Y_{t+\Delta} \left(1 - \theta_{1,t+\Delta} \mu_{t+\Delta}^{\theta_2}\right) \nabla_T \Omega\left(T_t\right)\right] \\ &= \sum_{i=1}^{\infty} \left(\beta_\Delta \Phi_T^\top\right)^i s_{t+\Delta i} Y_{t+\Delta i} \left(1 - \theta_{1,t+\Delta i} \mu_{t+\Delta i}^{\theta_2}\right) \nabla_T \Omega\left(T_{t+\Delta(i-1)}\right)\end{aligned}$$

- Bigger β raises the weight β_Δ^i on future marginal damages $\nabla_T \Omega$ in the shadow value of temperature $\lambda_{4,t} \rightarrow$ larger weight for future generations, SCC $\nearrow$

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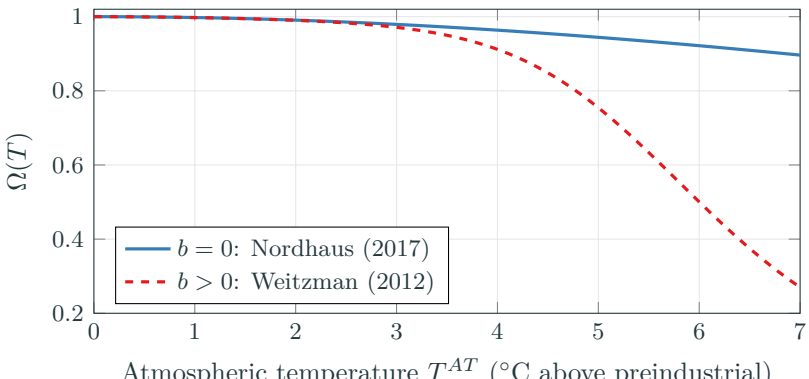
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The Damage Function and Tipping Points

Damage functions: Nordhaus vs. Weitzman

Share of output left after climate damages, $\Omega(T)$, up to 7°C:



Tipping points

- Tipping point are critical threshold beyond which a system reorganizes abruptly & irreversibly.
- Natural system tipping points well identified:
  - Antarctic & Artic ice sheet disintegration: complete disintegration (at +10°C) would raise the global sea levels by 53.3 metres;
  - Amazon Rainforest destruction: a warming planet →rainforest may transform into a dry savanna landscape;
  - Sibera's Permafrost thaw: unfreezing permafrost would release methane $\nearrow T°C$,
  - etc.
- Tipping points are possible at today's global warming of just over 1 °C (1.8 °F) above preindustrial times, and highly probable above 2 °C (3.6 °F) of global warming.

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$$\Omega\left(T_t\right) = 1/(1 + a_2\left(T_t^{AT}\right)^2 + bT_t^c)$$

where b and c are set to match potential losses as T grows: $b = 5.0703e-06$, $c = 6.754$.

	$\Omega(1)$	$\Omega(3)$	$\Omega(5)$
$b = 0$ - Nordhaus (2017)	0.9976	0.9792	0.9443
$b > 0$ - Weitzman (2012)	0.9976	0.9712	0.7544

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- Get back to FOC:

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$$\lambda_{4,t}^{AT} = \sum_{i=1}^{\infty} \left(\beta_{\Delta} \Phi_T^{\top}\right)^i s_{t+\Delta i} Y_{t+\Delta i} \left(1 - \theta_{1,t+\Delta i} \mu_{t+\Delta i}^{\theta_2}\right) \nabla_T \Omega \left(T_{t+\Delta(i-1)}\right)$$

- $\nabla_T \Omega \left(T_{t+\Delta(i-1)}\right)$ much larger under tipping climate, thus rising the Social Cost of Carbon and in turn the carbon tax.

Estimating damages: global vs. local temperature

Empirical damages: global vs. local temperature

- The damage function is the weakest link of IAMs: a_2 is calibrated, not estimated → can we **estimate** it from macro data?
- Bilal and Känzig (2026): shift the focus from **local** to **global** temperature:
  1. a 1°C global warming lowers world GDP by **over 20%** in the long run, an order of magnitude more than with country-level temperature;
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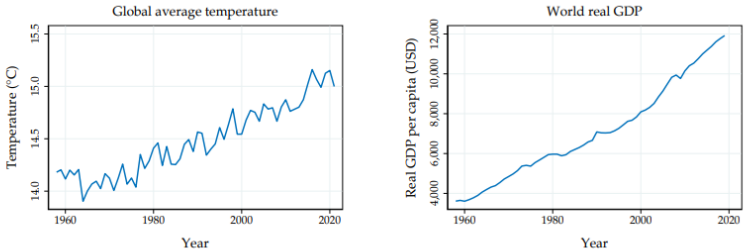
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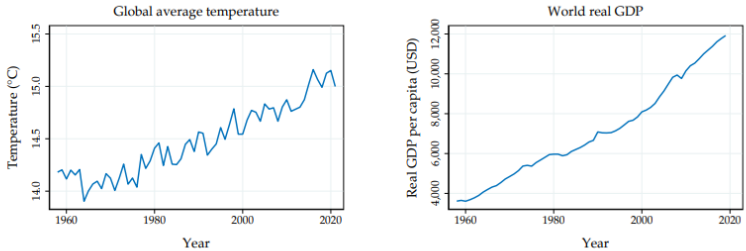
Global temperature and world GDP: two trending series



Notes: Global average temperature (including sea surface) from NOAA, world real GDP from PWT

- Both global temperature and world GDP **trend upward**.
- Regressing one on the other → risk of **spurious correlation**.
- Solution: focus on **temperature shocks**, i.e. transitory deviations around the trend.

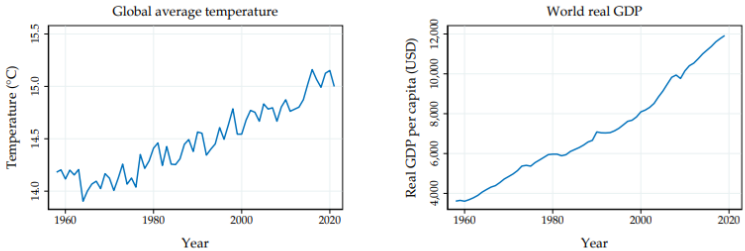
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Measuring temperature shocks

- Following **Hamilton (2018)**: the shock is the **forecast error** of temperature:

$$\widehat{T}^{\text{shock}}_{t+h} = T_{t+h} - \left(\hat{\beta}_0 + \hat{\beta}_1 T_t + \cdots + \hat{\beta}_p T_{t-p+1}\right)$$

- Sources of variation around the trend:
  1. exogenous drivers: solar cycles, volcanic eruptions;
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- $h = 2$, $p = 2$ to capture **persistent** climate phenomena (e.g. El Niño); results robust to other specifications.

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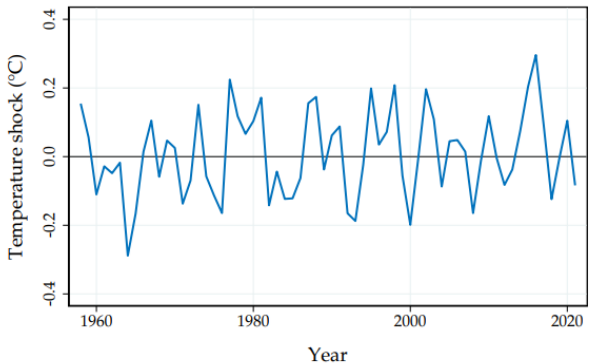
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Exogenous temperature shocks



Estimating the effects of global temperature shocks

- Dynamic causal effects estimated by **local projections** (Jordà, 2005):

$$y_{t+h} - y_{t-1} = \alpha_h + \theta_h T_t^{\text{shock}} + \mathbf{x}_t' \boldsymbol{\beta}_h + \varepsilon_{t+h}$$

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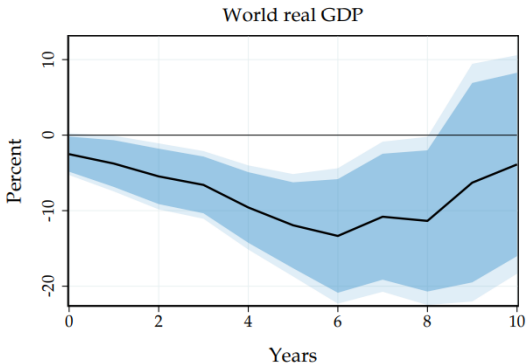
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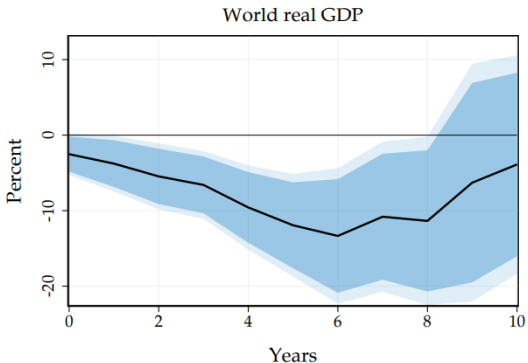
Effects of global temperature shocks



Notes: 90% and 95% confidence bands, robust standard errors.

- Significant and **persistent** effect.
- After a 1°C shock:
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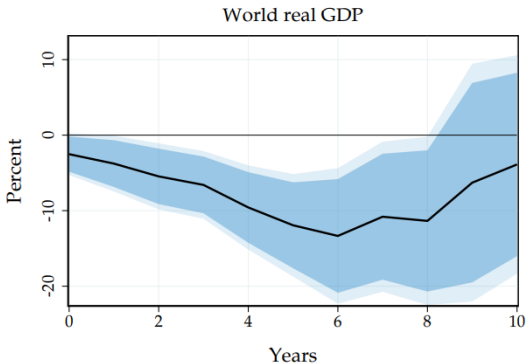
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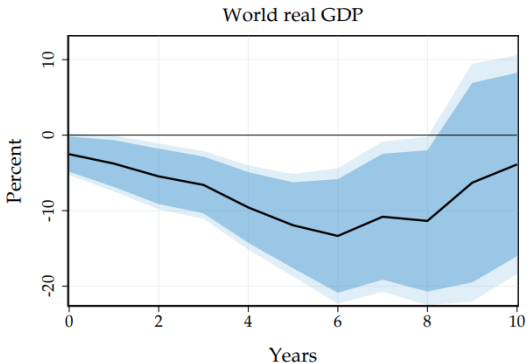
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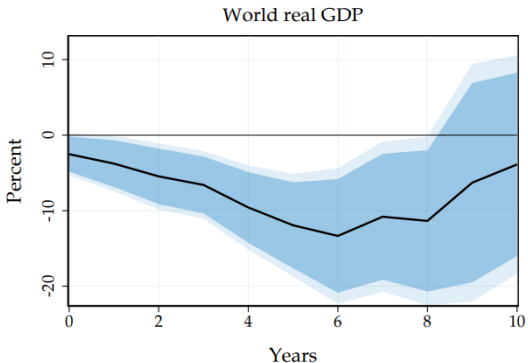
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A climate–economy panel

- **173 countries**, from 1960 (back to 1900 for some).
- Economic data: PWT and JST Macrohistory (GDP, population, capital, investment, productivity); temperature: Berkeley Earth; extreme events: ISIMIP gridded data.
- **Panel local projections** (Jordà et al., 2020):

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- Allows comparing responses to **global** and **local** temperature shocks.

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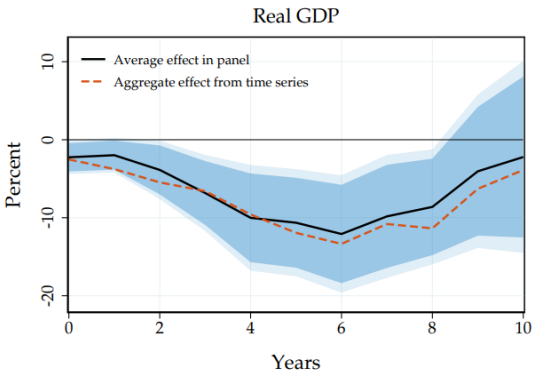
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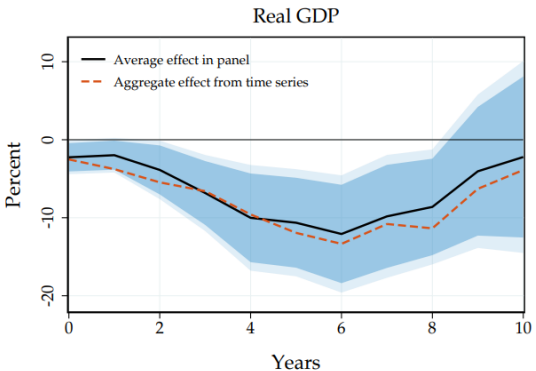
Global temperature shocks in the panel



Notes: 90% and 95% bands, Driscoll–Kraay standard errors.

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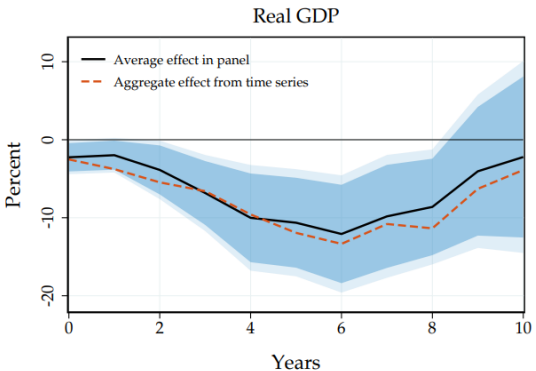
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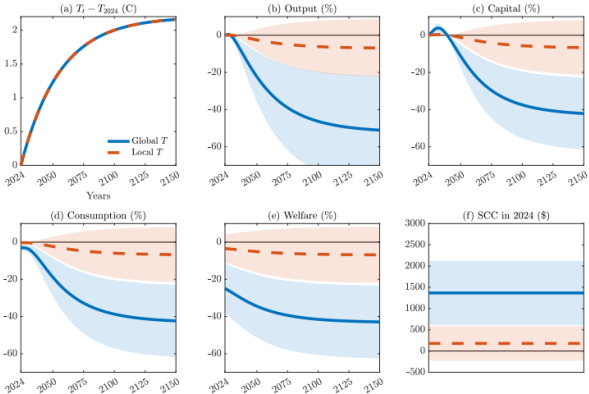
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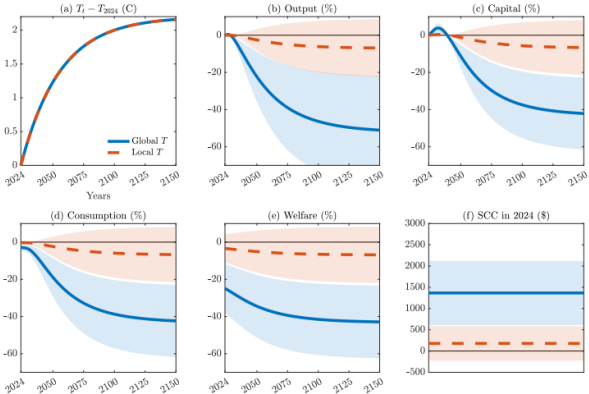
Implications for IAMs



Notes: simulated impact of global vs. local temperature damages, model calibrated in 2024.

- **Global** damages: in 2100, C and Y -40%; welfare loss 25%; SCC $= \$1\,367/\text{tCO}_2$.
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- Same model, different **damage evidence** $\rightarrow$ SCC $\times 7$: aggressive decarbonization becomes optimal.

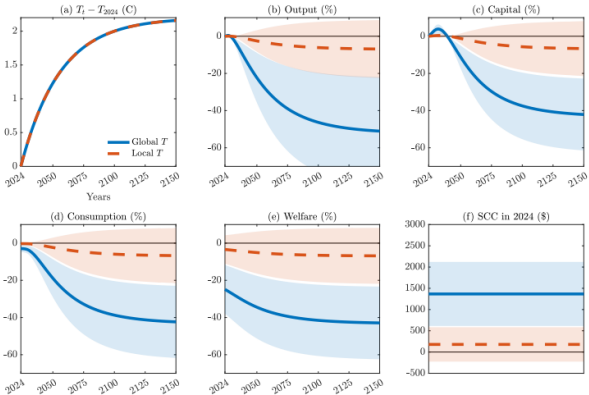
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Recap

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Thank you!

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